Open-Source Continuous-Culture Bioreactors: A Correction, a Comparison, and a Two-Instrument Proposal

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Prepared with AI assistance (Claude, Anthropic).
Source repository: `https://github.com/m9h/biopunk-bioreactor`

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Abstract

Low-cost open-source continuous-culture devices — turbidostats, chemostats, and morbidostats — have proliferated, but they are frequently compared as though they compete for a single role. They do not. We argue that the field serves *three* distinct purposes: in-situ *characterisation* of a culture held in a defined state; automated *evolution* that must produce archived, replicated evidence; and turnkey *scale-out* operation. We first correct a set of verifiable errors and overstatements in the recent OpenEvo preprint [2], including a self-contradictory growth-rate figure and a false claim about a competitor's licensing. We then place six platforms — Chi.Bio, OpenEvo, Pioreactor, eVOLVER, EVE, and the Toprak morbidostat — in a like-for-like matrix across fifteen attributes, and give per-platform strengths and limitations. Finally we propose that the best system is not one merged machine but *two* purpose-built instruments sharing a component ecosystem, and we note the strong educational role these devices already play. Every factual claim is tagged with its confidence level; specifications not verified against a primary source are flagged as such.

1 Introduction

Few organisms have shaped biology as thoroughly as brewer's yeast. The controlled fermentation of *Saccharomyces cerevisiae* is among the oldest of human technologies — beer, bread, and wine predate writing — and for most of that history it was practised as a continuous or serially propagated production process: a culture kept alive and productive across generations. When that ancient craft finally met experimental science it did not merely benefit from biology; it repeatedly *founded* it. Pasteur's demonstration in the 1850s that fermentation is the work of living yeast rather than spontaneous chemistry established microbiology and dealt the decisive blow to spontaneous generation [14]. Four decades later, Buchner's finding that a cell-free yeast extract still ferments sugar — that the transformation is catalysed by molecules, not a vital force — founded biochemistry. The glycolytic pathway, the spine of central metabolism, was mapped largely through fermentation; the first eukaryotic genome ever sequenced was that of *S. cerevisiae* [17]; and the genetics of the cell cycle, of protein secretion, and of homologous recombination were worked out in the same yeast that raises bread.

Two threads of that history bear directly on this paper. The first is *continuous culture*: the mid-twentieth-century formalisation of steady-state growth — Monod's growth kinetics [16] and the chemostat of Novick and Szilard [15] — turned the fermentation vat into a scientific instrument, a way to hold a population in a defined, unchanging environment and, crucially, to watch it mutate and adapt under sustained selection. The second is *production*: yeast remains the workhorse eukaryotic cell factory, from industrial ethanol to the first recombinant vaccine, and engineering better production strains is now a central problem of biotechnology. Continuous culture and directed evolution are where these two threads meet — an evolving population, held in continuous culture, pushed toward a phenotype it did not begin with. The open-source bioreactors reviewed here are the latest and cheapest instruments for exactly that meeting, democratising a program that once demanded Pasteur's laboratory or an industrial fermentation hall. It is fitting that OpenEvo's own optogenetic demonstration, and much of the continuous-evolution work discussed below, is again carried out in yeast.

The precision of an in-vivo biology experiment depends on isolating the subsystem of interest from the drift of a growing culture's environment. Continuous-culture devices address this by diluting a culture during growth: a turbidostat holds optical density (OD) at a set-point; a chemostat fixes the dilution rate and lets density float; a morbidostat titrates a drug to hold growth inhibition constant. Over the past decade a wave of open-source, low-cost instruments has made these techniques broadly accessible, and two in particular frame the present analysis: **Chi.Bio** [1], an all-in-one platform for in-situ measurement and manipulation, and **OpenEvo** [2], a recent preprint describing a cheap, buildable device for automated directed evolution.

These two are often read as competitors — both cost roughly $300, both are open — but they are optimised for opposite halves of the problem. Chi.Bio is fundamentally an *instrument*: it exists to measure a culture richly while holding it in a defined state. OpenEvo is fundamentally an *actuator*: it exists to push a culture through selection regimes, unattended, for weeks. Recognising this distinction — and a third purpose that a productised platform such as Pioreactor serves — reorganises the entire field and changes what "best" means.

This working paper has four aims. Section 2 sets out the three purposes and Section 3 the founding literature and a landmark demonstration of each. Section 4 corrects specific errors in the OpenEvo preprint, fairly and with proposed fixes. Section 5 compares six platforms attribute by attribute, and Section 7 argues for building two purpose-built instruments rather than one compromise machine. Section 8 documents the educational materials these platforms already serve. We are explicit throughout about what was verified against a primary source and what was not.

2 Three purposes, not one

Treating all continuous-culture devices as interchangeable obscures that they optimise for incompatible objectives.

`proto` — **Characterisation.** Hold a culture in a defined, unchanging state and measure it richly, in situ, in real time, closing feedback loops around it. The figure of merit is *signal density per reactor*, with a handful of reactors running different conditions in parallel. Chi.Bio is the exemplar: a chip spectrometer, a seven-colour LED for optogenetic actuation and fluorophore excitation, non-contact infrared thermometry, and proportional-integral-derivative plus model-predictive control that holds OD within about 2% of set-point [1].

`devo` — **Evolution.** Push a culture through selection regimes unattended for weeks, and produce *evidence* that adaptation occurred and which mutation caused it. The figures of merit are cheap chambers in numbers, systematic archiving, and long-run robustness. This is OpenEvo's target, and — as Section 4 argues — the half it does not yet fully reach.

`bior` — **Scale-out operation.** Run many reliable reactors now, with minimal build effort, a maintained software platform, cluster management, and support. The value is reproducibility and uptime, not measurement richness or archived evidence. Pioreactor (turnkey) and eVOLVER (maximally configurable) own this axis, which is orthogonal to the other two.

Read in order, the three trace a development lifecycle — `proto` (prototype and characterise a strain), `devo` (evolve it), `bior` (produce at scale) — which is why the tags are named for stages rather than numbered.

These purposes pull hardware in opposite directions — rich-and-few versus cheap-and-many, non-contact versus immersed-electrode-tolerant, defined-steady-state versus free-to-run-to-stationary. A device optimised for one is measurably worse at another. That tension is the organising principle of everything below.

3 Three lineages: the founding literature of each purpose

The three tags are not conveniences of this review. Each names a distinct research tradition with its own founding literature and its own landmark demonstration, and the traditions grew up separately over some seventy years. Setting them side by side makes the central point concrete: a single benchtop device now spans programs that were once the province of different laboratories and different instruments.

3.1 `proto` — characterisation

The characterisation tradition begins with the chemostat as a *measurement* instrument. Monod's growth kinetics and the Novick–Szilard chemostat [16, 15] made it possible to hold a population at a defined, unchanging physiological state and read its properties without the confounds of a changing batch environment. The decisive modern extension was to close the loop — to let a computer measure a living culture and act on it in real time; its founding demonstration is Milias-Argeitis and colleagues' *in silico* feedback, in which a controller regulated a gene-expression circuit in vivo by continuously measuring output and adjusting an input [18]. Chi.Bio [1] later folded sensing, actuation, and control into a single low-cost reactor. *Landmark demonstration:* automated optogenetic feedback control that held both gene expression and growth rate at prescribed set-points in continuous *E. coli* culture, tracking dynamic reference profiles and rejecting nutrient and temperature perturbations — a bioreactor regulating a cell's internal state the way a controller regulates an industrial plant [19].

3.2 `devo` — evolution

The evolution tradition begins where continuous culture was first turned on selection: Novick and Szilard used their chemostat not only to hold cells at steady state but to watch spontaneous mutations sweep a bacterial population [20]. It was systematised into experimental evolution by Lenski's Long-Term Evolution Experiment, whose founding paper followed twelve *E. coli* populations through their first 2,000 generations and established the discipline's central apparatus — replicate lineages and a frozen archive [21]. For molecules rather than whole organisms, the parallel founding is phage-assisted continuous evolution [12]. *Landmark demonstration:* after roughly 31,500 generations one LTEE population evolved the aerobic use of citrate — a trait so unusual it helps define the species — a result that could be established, and its historical contingency proven by "replaying" the tape from frozen ancestors, only because the lineages had been replicated and archived [8].

3.3 `bior` — scale-out

The scale-out tradition begins with continuous culture recast as a *production* technology. Herbert, Elsworth, and Telling's 1956 theoretical-and-experimental study established the quantitative basis for running continuous culture as an industrial process and described an operating pilot plant [22]. Its modern form is parallelisation — many small reactors run at once — from multiplexed chemostat arrays for high-throughput strain characterisation [23] to the open, automated sixteen-vial eVOLVER [3]. *Landmark demonstration:* eVOLVER-hosted phage-assisted continuous evolution, in which running many lagoons in parallel — the throughput that scale-out exists to provide — enabled the evolution of compact Cas9 variants with new PAM specificities [13].

These three lineages — physiological measurement, experimental evolution, and production — matured in different laboratories with different apparatus. What is new is not any one of them but that a device costing a few hundred dollars can now, in principle, serve all three at once; the argument of this paper is precisely that it should not try to.

3.4 An intersection: CRISPR and continuous evolution

The `bior` landmark above is itself a CRISPR result, and that is no accident: continuous culture and CRISPR are entangled in three directions. *These tools evolve CRISPR.* Phage-assisted continuous evolution, run in continuous culture, is now a primary engine for improving Cas enzymes — the broad-PAM, high-specificity xCas9 was evolved by PACE [24], cytosine base editors were evolved for expanded target compatibility and activity [25], and the compact Cas9 variants of the landmark itself came from scaling PACE onto eVOLVER [13]. Many of the CRISPR reagents now in routine use were, in part, made in bioreactors of this kind. *CRISPR is used inside these tools.* Pooled genetic libraries — the substrate of CRISPR screens — can be maintained and analysed in continuous culture [26], and Cas-based selection, linking editing or interference to survival, is a natural way to impose the growth-coupled selection that an evolution machine needs. *And CRISPR may help close the apparatus's hardest gap.* The archiving problem of Section 6.1 — the need for a frozen fossil record — has a molecular alternative in CRISPR-based recorders that write lineage and event history into a cell's own genome; whether an in-cell ledger can substitute for a physical archive on a cheap open platform is an open and appealing question.

4 Corrections to the OpenEvo preprint

OpenEvo is a genuine contribution and the corrections here are not a dismissal. Its real strengths include a peak-to-peak growth readout tied to hardware dilution events (which sidesteps local-maxima noise); a 940 nm OD path that avoids optogenetic and ambient overlap, with an inverse model holding $R^2 \geq 0.98$ out to $\mathrm{OD}_{600} = 6.0$; a closed autoclavable vessel with better sterility than Chi.Bio's open tube; a 4,000-hour remote run demonstrating robustness; and an illustrated assembly manual validated by having an inexperienced undergraduate build a working unit. It is the cheapest buildable three-media selection cycler in the field. The issues below are correctable and concern framing and analysis more than hardware.

4.1 Verifiable errors

[error] **The headline growth-rate figure contradicts itself.** The abstract states evolved cells "grew 55% faster than wild-type at 12% salt"; the results state "a doubling time of 4.9 hours

versus 7.6 hours (a 36% increase in growth rate).” Both describe the same data, but the two numbers are different quantities. A doubling time of $7.6 \to 4.9\,\mathrm{h}$ is a 36% *reduction in doubling time* yet a 55% *increase in growth rate*, since rate $\propto$ 1/(doubling time) and $(1/4.9)/(1/7.6) = 1.55$. The abstract’s 55% is the correct growth-rate figure; the results sentence mislabels the 36% doubling-time reduction as a growth-rate increase. *Fix:* state both explicitly — “36% shorter doubling time, equivalently 55% higher growth rate.”

[error] **A false claim about a competitor.** The preprint states that Pioreactor is “available for purchase, but only the software is open source.” A public `Pioreactor/hardware` repository exists and the hardware is openly licensed under CC BY-SA 4.0 (per its `LICENSE` file) [6]. The claim is used to clear competitive space and is false. *Fix:* describe Pioreactor accurately as open-hardware-and-software, distinguished not by openness but by being a supported product with a fleet platform. (We note in passing that an earlier draft of this document itself misnamed the licence as CERN-OHL; a contributor corrected it against the primary `LICENSE` file — an illustration of the citation discipline this repository aims for.)

4.2 Framing and omission

[framing] **The nearest prior art is uncited.** Chi.Bio [1] is open hardware and software, roughly the same $300, performs in-situ measurement and continuous culture, and predates OpenEvo by six years. The preprint surveys eVOLVER, the Toprak morbidostat, and Pioreactor, then frames an unfilled gap while omitting the system that most directly occupies it, which makes the gap look larger than it is.

[framing] **An unsupported superlative.** “The most affordable modern continuous growth system available” is not established: Pioreactor ships turnkey at $329 with no build, and EVE builds a triplicate system for $115–200 [4]. *Fix:* qualify to “among the lowest-cost *buildable* systems with three-media selection.”

[framing] **The success is overstated.** The 9% salt step did not partially adapt — growth ceased, OD flat-lined, and the run terminated at cycle 152. The adaptation claim rests on a single hand-picked glycerol stock at cycle 145, assayed offline. The result is real, but the framing should foreground one successful selection step with a failed extension, not a ladder.

4.3 Methodological limitations

[method] $n = 1$ **forecloses causal attribution.** A single clone from a single population was sequenced. Without independent replicate lineages there is no convergence signal, and the reported single-nucleotide polymorphisms and two large deletions cannot be attributed to the low-salt adaptation. This is the field’s central inference requirement — the Long-Term Evolution Experiment runs twelve populations for exactly this reason [8] — and it is the most consequential limitation.

[method] **The wrong variant caller.** Structural variants were called with “Geneious’s default mapper with low-medium sensitivity for 5 iterations.” The key finding — a $\sim$110 kb pHV4 deletion with “primarily transposon-associated elements” — is IS-mediated rearrangement, precisely the case a default short-read mapper handles worst. `breseq` exists for microbial-evolution resequencing including IS mobilisation [10], and its author is cited five times in the preprint. *Fix:* re-call with `breseq` and confirm the pHV4 structure with long reads.

[method] **No systematic archiving.** The preprint cites Barrick et al. — “. . . Archiving Populations, and Measuring Fitness. . . ” [7] — as motivation, then implements neither archiving nor fitness measurement in the automated system. Head-to-head ancestor-versus-descendant

Table 1: Six open continuous-culture platforms, attribute by attribute. Green marks a notable strength, red a notable weakness, amber a qualified case.

Attribute	Chi.Bio	OpenEvo	Pioreactor	eVOLVER	EVE
Primary purpose	proto	devo	bior	bior	devo
Reference	PLoS Biol '20	bioRxiv '26	none	Nat Biot '18	eLife '22
Parallelism	8 / computer	1 chamber	cluster	16 vials	≥3 reps
Working vol.	12–25 mL	∼20 mL	15–30 mL	∼40 mL	∼12 mL
OD source	650 nm laser	940 nm IR	900 nm IR	LED/PD	LED/PD
Rich sensing	spectrometer	OD only	plugin	config.	OD only
Contact	non-contact	closed	contact	contact	contact
Sterility	open air	closed, autocl.	autocl. vial	autocl. vial	3D-print
Selection	2+2 pumps	3 media+waste	pumps	config.	drug line
Control law	PID + MPC	threshold	Kalman GR	config.	threshold
Archiving	none	manual 1×	none	none	none
Open HW/SW	both	both	both	both	both
Cost/unit	\$300+PCB	∼\$300	\$329–389	<\$2k/16	\$115–200
Build effort	PCB fab	guided	turnkey	machine shop	moderate
Independent use	ReacSight	too new	community	ePACE	teaching

competition, the gold standard the cited work describes, is impossible without a time-series archive.

5 A like-for-like comparison

Table 1 places six platforms across fifteen attributes. Purpose codes are proto (characterise), devo (evolve), bior (scale-out). Cells reflect each platform as published or shipped, not what a motivated builder could add; "independent use" means peer-reviewed work by groups other than the originators.

The Toprak morbidostat [5] is omitted from the table for width but belongs to purpose devo: it is the canonical antibiotic-resistance device (working volume ∼12 mL), titrating a drug to constant inhibition, widely adapted but a dated bespoke build rather than a maintained platform.

Two patterns are worth naming. First, *every platform lacks archiving* — the single most important capability for turning an evolution run into evidence, and a clear gap for the whole field. Second, the only platform without a reference paper is Pioreactor, which is also the only pure bior product; its credibility is commercial and community rather than bibliographic, a genuine consideration when an instrument must be cited.

6 Why the evolution slot has no champion

Two of the three purposes have clear champions — Chi.Bio for `proto`, Pioreactor and eVOLVER for `bior` — but the `devo` slot is conspicuously empty, and OpenEvo, the one device built expressly for it, does not fill it. The reason is definitional. `devo` is not "push a culture through a selection regime" but "produce an archived, replicated, sequenced record that establishes that adaptation occurred and identifies the causal mutation." A champion must deliver the purpose's output, and OpenEvo delivers the run, not the evidence.

The shortfall is exactly the set of limitations catalogued in Section 4. With a single chamber and $n = 1$ there are no replicate lineages and hence no convergence signal, so the observed mutations cannot be attributed to selection rather than to drift or hitchhiking [8]. With no systematic archiving there is no frozen fossil record, and without one the ancestor-versus-descendant fitness competition that makes evolution quantitative is impossible [7]. And the single structural variant reported was called with a default short-read mapper rather than `breseq` [10]. OpenEvo is, in short, an excellent actuator with no evidentiary apparatus attached — the best pipette in the room with no freezer beside it.

Two further considerations keep the title vacant. The first is adoption: a champion earns standing through independent use by groups other than its originators, and the incumbent champions have it. Chi.Bio has been taken up by *ReacSight*, a framework that couples open bioreactor arrays — Chi.Bio among them — to an automated flow cytometer and a liquid-handling robot, so that experiments run *reactively*: each measurement feeds back to change the next actuation, turning a passive array into a closed-loop platform [11]. eVOLVER, for its part, has become a substrate for continuous directed evolution: it has been used to host *phage-assisted continuous evolution* (PACE) — the method of Esvelt, Carlson, and Liu in which a desired molecular activity is linked to the propagation of an M13 bacteriophage, so that proteins evolve tens of generations per day with minimal human intervention [12] — run across many parallel lagoons rather than the single bespoke apparatus of the original, a throughput increase that directly enabled campaigns such as the evolution of compact Cas9 variants [13]. This automated, eVOLVER-hosted form is what the community means by *ePACE*. OpenEvo, days old at this writing, has no independent use yet.

The second consideration is more pointed: the remedy does not obviously return to OpenEvo. What would constitute a `devo` champion — of order eight replicate chambers under one coordinating interface, an automated chilled archiver, and a `breseq` pipeline — most closely resembles Pioreactor's existing leader/worker cluster with an archiving plug-in, not a scaled-up OpenEvo. OpenEvo's durable advantage then narrows to its three-media selection fluidics and its closed autoclavable vessel, both of which are graftable onto another chassis.

None of this diminishes what OpenEvo is: the cheapest buildable three-media selection cycler in the field, with the best assembly manual among the six platforms, and the leading *candidate* for the role. But candidate is not champion, and the distance between them is precisely archiving, replication, and correct variant calling. Close those three and the slot is filled.

6.1 What the actuator must be connected to

The corollary of defining `devo` by evidence is that a champion is not a better actuator but an actuator wired into an evidentiary pipeline. OpenEvo already performs the actuation — controlled dilution, three-media selection, unattended runs of thousands of hours. What it lacks is everything *downstream* of the culture vessel. Five connections turn the actuator into a champion, and only one of them is new hardware.

1. A replicate manifold. The actuator must be one of many identical chambers under a single coordinating controller, so that a campaign runs of order eight or more independent lineages in parallel rather than one. Convergence across those lineages — the same locus mutated repeatedly — is the signal that a mutation was selected rather than drifting [8], and it cannot be reconstructed after the fact from a single run. This connection is a fleet interface, and the reference implementation already exists as Pioreactor's leader/worker cluster; the actuator connects *to* it rather than reinventing it.

2. A cold archive. The vessel's outflow — which already carries culture out on every dilution — must connect to an automated chilled fraction collector that banks a glycerol stock at programmed generation intervals. This is the frozen fossil record, the single component whose absence most distinguishes every current open platform from the Long-Term Evolution Experiment [7], and the one genuinely new module the champion requires. It is inexpensive: a refrigerated carousel and a diverter valve on an existing pump line.

3. A sequencing and variant-calling pipeline. Archived samples must flow to whole-genome sequencing and be called with `breseq` [10], whose handling of insertion-sequence–mediated structural variation is exactly what a default mapper lacks. This connection is off-instrument — a sample-submission workflow and a compute pipeline — but it is part of the apparatus, because an evolved strain whose mutations are miscalled is not evidence.

4. A fitness-measurement loop. The archive closes back onto the actuator: reviving a banked ancestor and competing it head-to-head against a descendant, read out through the same optical-density hardware, is what converts "the population changed" into a measured selection coefficient. Without the archive there is nothing to revive; with it, the actuator's own sensors become the fitness assay, and no new instrument is needed.

5. A selection-design layer. Finally, the actuator must be connected to a principled choice of selection pressure. Naive growth selection on a production strain evolves the mutant that deletes the costly pathway [9]; the champion couples product formation to growth — or deploys OpenEvo's negative-selection media line as counterselection — before the run begins. This connection is upstream and computational, a genome-scale design step, but it determines whether the evidence the apparatus so carefully collects is evidence of anything worth having.

The shape of the argument is therefore that the `devo` champion is mostly *connective tissue*: one new hardware module (the archiver), one borrowed fleet interface, one off-instrument pipeline (sequencing plus `breseq`), one feedback loop that reuses existing sensors, and one upstream design step. The actuator itself stays cheap and simple — OpenEvo's genuine strength — and the championship is earned by what surrounds it, not by loading the vessel with sensors. This is the same conclusion the two-instrument proposal reaches from the opposite direction (Section 7): resist instrumenting the chamber, and invest in the apparatus around it.

7 A two-instrument proposal

Because the three purposes pull hardware in opposite directions, the strongest system is not one merged machine but two purpose-built instruments that share a component ecosystem.

7.1 The best characterisation instrument

Start from Chi.Bio; its architecture is already correct. The governing constraint is that *non-contact measurement is inviolable* — every subsystem but the heat plate reads through the vessel wall, which is why tubes hot-swap and sterilisation is trivial. Upgrades that respect this: dissolved-oxygen and pH optode spots read optically through the wall; a dispersive micro-spectrometer replacing eight fixed filter bands with a continuous 340–850 nm trace; and a sealed, gas-controlled lid to close Chi.Bio's open-to-atmosphere gap. Viable-cell-density by dielectric spectroscopy is explicitly *excluded* here because it requires immersed electrodes — it belongs on the other machine.

7.2 The best evolution machine

Start from OpenEvo; its virtues (cheap, closed, buildable, standalone) fit long unattended runs. But its two missing capabilities are not sensors. First, *replicate populations*: $n = 1$ is a failure of causal inference, so the build is eight cheap chambers under one coordinating interface, not one richly instrumented reactor. Second, *archiving*: an automated chilled fraction collector on the existing waste line, diverting a programmable fraction on a generation-count schedule into a chilled glycerol archive — the capability that makes fitness measurable, and, as far as we can determine, one nobody has built. A *batch-excursion* firmware mode — periodically suspending dilution to run to stationary phase — would additionally recover carrying capacity, a phenotype a turbidostat structurally cannot observe, and the very axis on which the OpenEvo cycle-145 population differed. The adaptive-laboratory-evolution literature adds a crucial caution: selection is for growth, and product formation costs growth, so growth-coupled strain design must precede selection [9].

7.3 Why two, not one

The sensor modalities sort themselves by contact. Optode DO/pH and dispersive spectrometry are optical and non-contact, and land on the characterisation instrument; conductivity, electrical resistance tomography, and dielectric viable-cell-density need immersed electrodes and land on the evolution machine, whose closed autoclavable vessel welcomes them. Almost no sensor is wanted by both and available to only one — the clearest evidence these are two instruments. What they *share* is an ecosystem: a common module footprint, reusable sensor-module designs, one firmware framework, and a shared calibration reference. Combine at the platform level, not the product level.

8 Educational role

These platforms are teaching instruments as much as research tools, and because one device spans microbiology, electronics, firmware, control theory, and data analysis, a single unit serves biology, engineering, and computing classes at once. EVE is the education-first exemplar: its paper is explicitly titled "...and its educational use," it builds for under \$200, and it was validated in a real classroom evolving replicate *E. coli* under chloramphenicol [4]. Pioreactor maintains an educators portal and lesson concepts spanning OD-by-scattering through evolution. OpenEvo's assembly manual doubles as a build-it-yourself course and anchored an international hackathon in which students remotely controlled a device on another continent. Chi.Bio's own documentation offers its core features as a stand-alone teaching platform on which students can observe fluorescence by eye. Notably, the teaching materials inherit the research blind spots: archiving, replicate-based inference, and carrying capacity are as under-taught as they are under-implemented — an opportunity for a device-agnostic curriculum.

9 Discussion and limitations

This is a reader's audit, not peer review, and confidence is uneven. Full texts of Chi.Bio and OpenEvo were read directly. Pioreactor specifications, pricing, hardware licensing (CC BY-SA 4.0, verified against the primary `LICENSE` file), and the absence of a reference paper were confirmed from primary or product sources, as were the headline specifications of eVOLVER and EVE and the working volume of the Toprak morbidostat. Component datasheets underpinning the proposed evolution machine (spectrometer, spectral sensor, and impedance front-ends) were verified in a companion pass. Some adoption specifics and a small number of vendor register-level details remain unverified at page level and are flagged as such in the repository's `SOURCES.md`. No claim here should be budgeted against without confirmation at its primary source.

10 Conclusion

The open-bioreactor field is best understood as serving three purposes, not one contest. Chi.Bio owns characterisation; Pioreactor and eVOLVER own scale-out; and evolution — the purpose with the most devices chasing it — has no clear champion, because no platform yet archives or replicates as the science requires. OpenEvo's real contribution is the cheapest buildable selection cycler with a genuinely excellent manual; its preprint's errors of framing and analysis are correctable and worth correcting. The best path forward is two purpose-built instruments sharing a parts ecosystem, with archiving and replication treated as first-class — and with the educational mission, already strong, extended by a curriculum that teaches the methodological rigour the hardware has so far left out.

Source data, the full comparison matrix, per-platform pros and cons, a datasheet-verification pass, an educational-materials guide, and this document's LaTeX sources are maintained at https://github.com/m9h/biopunk-bioreactor *under CC BY 4.0. Corrections with citations are welcome.*

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